

Frame It as Cases: Work That Speaks for Itself

Samar Jamal | B.Tech AI & ML Student | Amazon ML Summer School Trainee

Proof Statement: As a B.Tech AI & ML student and Amazon ML Summer School trainee, I build and evaluate leak-free, production-ready machine learning pipelines on real data. This portfolio exists for a Senior Machine Learning Engineering Lead at an AI-first tech company who values evaluation rigor, clean PyTorch code, and benchmark transparency over generic resume buzzwords. Primary action: schedule a 15-minute technical screening call with me.

Case Study 1: MediScan AI — RAG Healthcare Platform

Out-of-the-box LLMs hallucinate medical advice. Built a full-stack RAG system (TypeScript, Node, Pinecone, Gemini API). Designed contextual query rewriting and similarity thresholding ($\cos \text{sim} < 0.72 \rightarrow \text{'Insufficient context'}$) to eliminate hallucinations. Latency $< 1.2\text{s}$.

Before vs After: Before: 'State-of-the-art revolutionary AI platform...'

After: 'Built full-stack RAG medical intelligence platform using TypeScript, Pinecone, and Gemini API with 0.72 similarity thresholding to eliminate hallucinations.'

Case Study 2: AI Disaster Response Coordinator — RL Environment

Disaster response requires dynamic priority triage under supply constraints. Built OpenEnv-compliant RL simulation environment. Benchmarked Qwen2.5-72B-Instruct against priority-dispatch heuristics, achieving an 0.884 weighted score across Easy, Medium, and Hard tiers.

Before vs After: Before: 'Groundbreaking AI disaster management system...'

After: 'Built OpenEnv RL environment for disaster allocation, achieving 0.884 weighted score with Qwen2.5-72B.'

Case Study 3: Scene Classification with EfficientNet-B2

Training 60K+ image classifier (SUN397, 397 classes) suffers from GPU memory bottlenecks. Fine-tuned EfficientNet-B2 with PyTorch & CUDA AMP, cutting GPU memory by 45% and epoch time from 18m to 9.5m. Achieved 75% macro F1-score with Test-Time Augmentation.

Before vs After: Before: 'High-performance deep learning model with industry-leading accuracy...'

After: 'Fine-tuned EfficientNet-B2 on SUN397 using PyTorch CUDA AMP, achieving 75% macro F1 while cutting training time by 47%.'

Case Study 4: SenseLink — Edge-AI Assistive System on ESP32

Visually/hearing impaired users require low-latency navigation without cloud dependency. Built edge hardware on ESP32 combining quantized YOLO object detection and live captioning for real-time auditory/visual navigation alerts.

Before vs After: Before: 'Futuristic IoT accessibility wearable...'

After: 'Built edge-AI assistive system on ESP32 combining quantized YOLO detection and on-device captioning.'

Bio & Contact

Bio: B.Tech AI & ML student at GEHU & Amazon ML Summer School trainee specializing in leak-free ML pipelines, PyTorch computer vision, and full-stack AI systems.

Links: github.com/Samarjamal326 | linkedin.com/in/samar-jamal | Samarjamal326@gmail.com

Call to Action: Schedule a 15-minute technical screening call with me.